WEEKLY LESSON PLAN · 2026-27

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 · Advanced · Senior · Mr. Gavaldon · Da Vinci School
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SEMESTER 1
Week 2

MONDAY

Day 6 - Launch Fusion + First Sketch

Aug 17

OBJECTIVE
Students will construct a fully-constrained, dimensioned sketch in Fusion 360 by applying geometric constraints and parametric dimensions to drive the sketch to zero degrees of freedom, and will explain how design intent lets one parameter change propagate through the model.

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - industry-standard CAD (this is the shared Semester-1 Fusion foundation)

ELPS / EB SUPPORT
ELPS/EB supports at-level: every constraint symbol shown on-screen and on a station card (VC, DP); pre-teach parametric / constraint / degree-of-freedom with a picture wall before the build (PV); directions restated in short numbered steps (CD, RS); fully-defined finishers coach a partner in Spanish or English (PN); wait time after the DOF question (WT).

AGENDA · 45 MIN

- **5′** Bell-work: Define parametric CAD in one sentence; list two things a driven dimension does that a freehand line cannot.
- **10′** Interface + parametric philosophy: browser, timeline, sketch palette; watch Part 1 intro
- **8′** Concept + worked example: geometric constraints and degrees of freedom (drive DOF to 0)
- **15′** Guided build: fully-constrain a dimensioned bracket profile anchored to the origin; name parameters
- **5′** Constraint & DOF audit worksheet, then save to the cloud
- **2′** Exit ticket: why must a profile be fully defined before you extrude?

SLIDES / ACTIVITY
Parametric Sketching in Fusion 360

NOTEBOOK
Sketch the Fusion screen and label the toolbar, sketch button, and timeline.

TURN IN
Sketch Constraint & Degrees-of-Freedom Audit

TUESDAY Day 7 - Sketch to Part

Aug 18

OBJECTIVE

Students will create a solid body by extruding a closed profile, apply fillets and chamfers with engineering justification (stress concentration and assembly), and compute the part's volume and mass from its geometry and material density, then reconcile the hand calculation with Fusion's reported mass properties.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - CAD - solid modeling (shared Fusion foundation)

ELPS / EB SUPPORT

ELPS/EB supports at-level: live demo of each feature before work time with the icon shown (VC, RS); one-page step card per station (CD); pre-teach extrude / fillet / chamfer / density with a labeled diagram (PV, DP); the mass calculation modeled once, numbers left on the board (RS); partners check each other's arithmetic (PN); wait time on the exit question (WT).

AGENDA · 45 MIN

- 5′ Bell-work: What is the difference between a fillet and a chamfer, and when would you use each?
- 10′ Extrude operations (join / cut / intersect, symmetric, taper) and the feature timeline
- 8′ Fillets, chamfers, stress concentration, and mass properties - worked example
- 15′ Build: extrude your profile to a solid, add a fillet and a chamfer, assign a material, read mass properties
- 5′ Mass-properties calculation worksheet
- 2′ Exit ticket: your hand-computed mass vs. Fusion's reported mass - do they agree, and why or why not?

SLIDES / ACTIVITY

From Profile to Parametric Solid

NOTEBOOK

List the 3 steps you used to turn a sketch into a solid.

TURN IN

Mass Properties & Feature Plan

WEDNESDAY Day 8 - Model a Simple Object

Aug 19

OBJECTIVE

Students will select a modeling target using a weighted decision matrix, decompose it into CAD features, model it to caliper-measured dimensions, and verify dimensional accuracy by executing a measurement plan and computing the percent error between the physical object and the CAD model.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - CAD - model a real object to dimension (shared Fusion foundation)

ELPS / EB SUPPORT

ELPS/EB supports at-level: the decision matrix modeled column-by-column on the board (VC, RS); pre-teach criteria / weight / percent error / verification (PV, DP); caliper technique demonstrated then practiced with wait time (WT); step card for the measure-model-compare loop (CD); partners measure and cross-check three times (PN).

AGENDA · 45 MIN

- 5′ Bell-work: List four criteria you would use to decide which object is worth modeling this period.
- 8′ Concept selection with a weighted decision matrix - worked example
- 7′ Reverse engineering: measuring to tolerance with calipers; building a verification plan
- 18′ Model your chosen object to measured dimensions; log measured vs. modeled for each feature
- 5′ Complete the decision matrix and the verification data table
- 2′ Exit ticket: which feature had the largest percent error, and how will you iterate?

SLIDES / ACTIVITY

Reverse-Engineering a Real Part

NOTEBOOK

Sketch your object with real dimensions before you model it.

TURN IN

Concept-Selection Weighted Decision Matrix, Reverse-Engineering Verification: Test Plan + Data

THURSDAY

Day 9 - Design Choices + Prep to Present

Aug 20

OBJECTIVE
Students will prepare a technical design brief that justifies their design decisions with evidence - citing their decision matrix result, dimensional-verification data, and mass properties, and naming at least one explicit trade-off - using precise engineering vocabulary and a structured claim-evidence-reasoning format.

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - technical communication - presenting a design & its rationale

ELPS / EB SUPPORT
ELPS/EB supports at-level: sentence frames that still demand a number ('I chose ___ because the data showed ___') (CD, RS); pre-teach rationale / trade-off / specification (PV); model brief shown and annotated (VC, DP); rehearse in pairs before whole-class with wait time (PN, WT).

AGENDA · 45 MIN
- 5′ Bell-work: Name one trade-off in your model - what you improved and what it cost.
- 10′ What makes a rationale 'engineering' vs. opinion: claim, evidence, reasoning, and trade-offs
- 8′ Trade-off analysis and the technical spec sheet - worked example
- 15′ Build your ~90-second technical brief and spec sheet; cite three real numbers
- 5′ Rehearse once with your team using the self-check rubric
- 2′ Exit ticket: state your strongest evidence-backed claim in one sentence.

SLIDES / ACTIVITY
Defending a Design with Evidence

NOTEBOOK
Write your 1-minute pitch outline: what it is, why these choices, one thing you'd improve.

TURN IN
Technical Design Brief + Trade-off Analysis

FRIDAY

Day 10 - First Team Presentations

Aug 21

OBJECTIVE
Students will deliver a ~90-second technical design brief defending their model with quantitative evidence, and will evaluate peers with a weighted rubric - computing a weighted score out of 4 and writing specific, evidence-based feedback tied to a criterion and a number.

TEKS
§Eng Design & Presentation TEKS (draft - verify official section) - deliver a technical presentation; give & receive feedback

ELPS / EB SUPPORT
ELPS/EB supports at-level: rubric levels shown with a worked example score so evaluation isn't language-gated (VC, DP, CD); feedback frames that force a criterion + number ('Your ___ scored high because ___') (RS); presenters use their rehearsed script (WT); peers may confer before scoring (PN).

AGENDA · 45 MIN
- 5′ Bell-work: What counts as evidence in a design defense? Give one example with a number.
- 6′ Calibrate the weighted evaluation rubric - what a 4 vs. a 2 looks like on each criterion
- 27′ Technical briefs - present (~90 s each) and score peers on the weighted rubric
- 5′ Tally weighted scores; write one glow and one grow tied to a criterion + number
- 2′ Exit ticket: name the ONE iteration you'll make based on the feedback you received.

SLIDES / ACTIVITY
Technical Briefs & Evidence-Based Critique

NOTEBOOK
Log your pitch (what went well, one thing to improve) + one piece of feedback you got.

TURN IN
Weighted Peer Evaluation + Iteration Log